Reflection — Learning Journey

Starting from scratch: concepts before tools

I started this lab without a clear mental model of how Tableau is structured. Before touching any data, I needed to understand what a worksheet actually is versus a dashboard versus a workbook — concepts that sound obvious but are genuinely confusing when you have never used the tool. The distinction that clicked first was thinking of worksheets like individual PowerPoint slides, and the dashboard as the layout that assembles them. The concept of a Project confused me initially because it appeared in documentation but not in Tableau Desktop — I later learned it only exists in Tableau Server and Cloud, which was a useful reminder that the same tool behaves differently depending on the environment.

Choosing and preparing the data

Rather than using the lab's synthetic evaluation dataset, I chose the World Happiness Report 2024 from Kaggle and enriched it with WHO Life Expectancy data. This immediately introduced a complexity the lab did not anticipate: the two datasets used different country name conventions. "Russian Federation" in WHO, "Russia" in the Happiness file. "Cote d'Ivoire" versus "Ivory Coast". These mismatches would have caused silent null values in Tableau — rows that simply disappear without any error message. Fixing them required going through both files in Excel and standardising 12 country names before importing anything. The lesson here was that data preparation is not a footnote — it is the foundation everything else rests on, and skipping it costs far more time later than doing it upfront.

The calculated field that would not calculate

One of my first real blocks came when creating the `Development Status` calculated field. The formula `IF [Developed] = "1" THEN "Developed" ELSE "Developing" END` threw a type error. The problem was that the WHO file stores `Developed` as a number (0 or 1), not as text — so comparing it to `"1"` in quotes caused a type mismatch. The fix was simply removing the quotes: `IF [Developed] = 1`. A small difference visually, but it revealed something important about how Tableau handles data types silently. The # icon versus Abc icon in the Data pane became a check I now do automatically before writing any conditional logic.

Too many trend lines

When I built the Health vs Happiness scatter plot and added a trend line, I got ten lines instead of one — one per world region, because Regional indicator was on the Color shelf. Tableau's default behaviour when a dimension is on Color is to draw a separate trend line per category. The fix was right-clicking any trend line → Edit Trend Lines → unticking "Show a trend line per color". This was a case where Tableau's automatic behaviour was technically correct but analytically wrong for what I was trying to show. I wanted one overall correlation line, not ten regional ones pulling in different directions across the canvas.

The dashboard filter chaos

Building the dashboard introduced the most layered set of challenges. The first was an empty "Filters..." placeholder box that appeared on my dashboard — I had dragged something incorrectly and a container with no content had landed on the canvas. Deleting it and rebuilding the filter bar using a Horizontal Layout Container with filter controls dragged in manually was the right approach, but understanding that filter controls in Tableau are "owned" by a specific sheet and need to be explicitly promoted to global scope was not intuitive. Every filter required the extra step of "Apply to Worksheets → All worksheets using this data source" — without that step, selecting a region only updated one chart while the others ignored it entirely.

The most frustrating moment came when applying the Development Status filter caused entire charts to go blank. The Score Distribution and Scores by Region worksheets — which contain only Happiness data — had no concept of a WHO field. When the filter was applied globally, Tableau tried to evaluate `[Developed]` for those rows, found nothing, returned null for every country, and emptied the chart. The solution was to scope Development Status to "Selected Worksheets" and apply it only to the Health vs Happiness scatter, which is the only chart that actually joins WHO data. This taught me a rule I will carry forward: only apply a filter to sheets that use the field the filter is derived from.

The filter that filtered out everything

A quieter but equally confusing moment was when the Score Bucket filter made the chart go blank. The filter dialog showed three values — High, Medium, Low — but the Summary line at the bottom read "Selected 0 of 3 values". All boxes were unticked, including (All), so the filter was actively excluding every row. Ticking (All) resolved it in two seconds, but understanding why it happened was important: Tableau filters default to nothing selected when first created, which means they exclude everything by default. The correct initial state for a dashboard filter is (All) ticked, letting users narrow from there rather than starting from empty.

What I would do differently

The biggest change I would make is to plan the cross-source filtering behaviour before building, not after. I spent time undoing filter scopes that I had set incorrectly because I applied "All worksheets" uniformly without thinking about which sheets actually used which data source. A simple table — filter name, source field, applicable sheets — written before touching the dashboard would have saved that back-and-forth. I would also spend more time on tooltip design. The default tooltips expose raw field names like "AVG(Ladder score)" and "SUM(GDP per capita (WHO))" rather than clean readable labels, which contradicts the non-technical audience principle I applied to everything else. That is the one element of the dashboard that still reads like a developer built it rather than a communicator.